

wear, it has been shown that stochastic resonance improves balance and gait.

Mechanical actuators, designed to be inserted into any type of footwear, are embedded into the insoles. A signal generator and a small battery are also integrated within the insole to provide sensory enhancement stimulation to



Researchers at Harvard University say this technology can modify gait to improve balance and even enhance agility for athletes

the user's feet. A smart phone app controls the stimulation level and can inform the user of device status such as stimulation and battery levels.

When used by elderly individuals, this sensory enhancement insole could potentially improve their sense of balance and walking stability, which is understood clinically to be related to the risk of falling. For diabetics, who suffer from peripheral neuropathy, this kind of device may increase the sensitivity of their feet and might eventually decrease the risk of ulceration.

LED lights to be used for tumour detection

According to a recent release from Nanoscale, an interdisciplinary team from Rice University, USA, has confirmed on the basis of thorough experimentation, that LED lights have an enormous potential to identify tumours, by shining the light on these. Such a conclusion was drawn using an instrument based approach, dubbed as spectral triangulation system, which is a customised version of the modern medical imaging system.

Team leader Bruce Weisman, along with his group, placed detection of carbon-nanotube marker tagged, antibody associated tumour cells as their main objective. As an explanation, they said that the absorption coefficient of short wave infrared light in tissues feature different wavelengths at different levels. Owing to such phenomena, light-spectra analysis was preferred as it could produce information related to the depth of tissues through which light has passed.

World's first 3D-printed motorcycle

From bridges to shoes to metal parts used in airplanes, 3D printing has revolutionised the ability to create lightweight and affordable parts. Recently, this innovative technology has helped launch the world's first 3D-printed motorcycle.



Light Rider uses bionic algorithms to define the organic frame structure using nature's design principles

Built by APWorks, a Germany based subsidiary of French aeronautics firm Airbus, the motorbike is powered by a six-kilowatt electrical engine that can zip from zero to 80kph in seconds. The motorcycle, dubbed Light Rider, features a frame that weighs 6kg, making the midsection of the craft roughly 30 per cent lighter than that of conventional all-electric motorcycles.

APWorks constructed Light Rider's frame by lacing together an intricate web of Scalmalloy, an aluminium-like alloy powder that forms a solid that is nearly as strong as titanium.

Biosensor to detect cancer, Alzheimer's, Parkinson's

Researchers have developed a biosensor that is capable of detecting molecules that lead to some types of cancer as well as neurodegenerative diseases. The device is basically a single-layer organic nanometer-scale transistor on a glass slide.

The biosensor contains the reduced form of the peptide glutathione (GSH), which reacts in a specific way when it comes into contact with the enzyme glutathione S-transferase (GST), linked to Parkinson's, Alzheimer's and breast cancer, among other diseases.

Findings show that the device could detect such molecules even when these are present at very low levels in the examined material, because of its nanometric sensitivity.